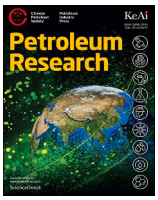
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Review Article

Advancing low-salinity waterflooding for improved oil recovery: Mechanisms, machine learning, and future directions

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ABSTRACT

The efficacy of low-salinity waterflooding for improving hydrocarbon recovery has garnered significant attention, owing to its greater efficiency in improving oil recovery compared to conventional high-salinity waterflooding. However, despite its widespread recognition, most studies have focused on the core size using core flooding and spontaneous imbibition. As a result, the understanding of this phenomenon remains limited, leading to uncertainty about the feasibility of implementing large-scale applications. This is primarily attributable to the obscurity of underlying mechanisms and the inconsistency of research results. Furthermore, conventional techniques for estimating oil recovery factors through low-salinity waterflooding have inherent limitations. These methods are time-consuming, require specific data and expertise, and lack built-in uncertainty quantification. Machine learning has gained significance in the petroleum sector in recent years due to its adaptability and its ability to manage large datasets efficiently. The integration of machine learning into low-salinity waterflooding offers a valuable opportunity to develop predictive models that effectively address constraints associated with conventional methodologies. Machine learning enables fast, accurate, and impartial estimation of oil recovery factors through advanced data-driven strategies. This study provides a systematic review of recent advances in the intelligent prediction of oil recovery factors using machine learning techniques in low-salinity waterflooding. It explores various machine learning models, their effectiveness in addressing existing challenges, and their potential to optimize low-salinity waterflooding performance in large-scale field applications.

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1. Introduction

The increasing demand for energy, coupled with the rapid decline in oil extraction from traditional methods, has compelled the petroleum sector to focus on the exploration and development of unconventional hydrocarbon reserves (Arain and Ridha, 2026). Despite the success of unconventional hydrocarbon resources, pursuing these sources has brought numerous challenges. Therefore, researchers have turned to enhanced oil recovery (EOR) techniques as a promising avenue. This was particularly for extracting residual oil from mature reservoirs after primary recovery. Following primary recovery, a significant volume of oil remains confined within the reservoir due to a reduction in initial reservoir pressure (Afifi et al., 2021; Deng et al., 2021; Ghasemi and Shafiei, 2022; Hussain et al., 2022; Khosravi et al., 2024; Mwakipunda et al., 2023). Employing various techniques to extract this specific proportion of oil may enhance output in current oilfields, ensuring a sufficient supply for future oil requirements.

Waterflooding techniques are the most viable method for improving oil recovery and producing residual oil from mature reservoirs after depleting primary recovery. Waterflooding is the injection of freshwater or brine to sustain reservoir pressure (Goolsby and Anderson, 1964; Rausch and Beaver, 1964). Low-salinity waterflooding (LSWF) is a recently advocated technique

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for improving oil recovery. This method involves altering or reducing the ionic composition and concentration of brines (Bartels et al., 2017a, 2017b). The brine introduced into LSWF exhibits reduced dilution and saline concentration compared to traditional high-salinity waterflooding (HSWF) processes. LSWF garnered significant interest after demonstrating a significant 20%–50% decrease in residual oil saturation (Aladasani et al., 2012; Webb et al., 2004). Introducing low-salinity fluids can alter the interactions among crude oil, brine, and rock (COBR) within the reservoir, thereby enhancing the recovery of residual oil (Agrawal et al., 2024; Arain et al., 2024b). These interactions may involve alterations in rock wettability and a decrease in the interfacial tension between oil and water (Al-Hadhrami and Blunt, 2001; Arain et al., 2024a, 2024b). LSWF's application has become increasingly significant owing to its straightforward injection process and advantageous economic factors (Chen et al., 2023; Farooqi et al., 2021; Lyu et al., 2022; Saeed et al., 2022; Sagbana et al., 2023).

Recently, there has been increasing interest in developing and utilising machine learning (ML) techniques across diverse applications in the oil and gas sector. The application of ML has been shown to be effective in several engineering fields. ML methodologies are increasingly being adopted to address a range of challenges across diverse sectors of the petroleum industry (Arain et al., 2022). Numerous sectors of the upstream petroleum industry, including reservoir engineering, drilling, and production engineering, have greatly benefited from efficient, cost-effective ML techniques. These methods can provide valuable insights and help address issues related to processing and interpreting large databases. Furthermore, it has been well recognised for its several benefits compared to the conventional empirical. The primary factors are their capacity and superior effectiveness in discerning the non-linear connections between high-dimensional input and output data, which are often not captured by empirical correlations (Farooqi et al., 2021; Otchere et al., 2021).

There is a substantial rise in the generation of experimental and field data across several fields. This has led to the application of ML methods to predict and enhance field performance. Despite the widespread utilisation of ML in EOR (Li et al., 2025), there is a scarcity of research investigating the efficiency of LSWF through the development of intelligent ML predictive models (Hidayat and Astsauri, 2022; Wang and Fu, 2018). The process of selecting the most appropriate IOR/EOR technique using experimental screening procedures and traditional reservoir modelling and simulation methodologies is expensive and time-consuming. An assemblage of predictive models, including the Artificial Neural Network (ANN), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), Extra Tree (ET), and the Least Squares Support Vector Machine (LSSVM), can be utilised in the estimation of LSWF potential (Kondori et al., 2021; Tatar et al., 2021). However, accuracy, reliability, and robustness are necessary for the developed ML methods to objectively evaluate LSWF performance.

A multitude of statistical metrics, including the coefficient of determination (R^2), mean squared error (MSE), and relative deviation, are employed to validate the developed models. The implementation of ML to predict LSWF performance for improved oil recovery (IOR) offers significant advantages, including broad applicability, high precision, and simplicity. However, despite the widespread utilisation of ML, there is a scarcity of research that has examined the efficiency of LSWF through the development of intelligent ML predictive models (Hidayat and Astsauri, 2022; Wang and Fu, 2018). Therefore, it is crucial to summarise and evaluate the current state of advancement in this topic to provide recommendations for practical applications and to suggest innovative directions and ideas for future investigation.

This study presents a comprehensive, integrative review of the fundamental mechanisms governing LSWF, while critically assessing the application of ML techniques for predicting oil recovery factors. It explains the key physicochemical interactions underlying LSWF and highlights the value of ML as a complementary approach for modelling the complex, nonlinear behaviour of reservoir systems beyond traditional mechanistic analyses. While previous studies have primarily focused on mechanistic interpretations derived from core-flooding and spontaneous imbibition tests, this review highlights how data-driven models can address challenges such as uncertainty quantification, model generalizability, and scalability. It consolidates recent advancements in machine learning algorithms, data preprocessing, and validation strategies specifically for LSWF, identifies key research gaps, and outlines future directions for hybrid physics-informed predictive frameworks. This work provides a clear roadmap for optimizing LSWF performance through intelligent, scalable, and reliable predictive models.

2. Mechanisms of low-salinity waterflooding

2.1. Fines migration

The fine migration mechanism was first introduced by Tang and Morrow (1999). The investigation was conducted using several Berea sandstone cores and postulates that low-salinity water may induce the separation of clay fragments, particularly kaolinite, thereby enhancing the wettability of clay minerals toward a more water-wet state. Following the injection of low-salinity water into the formation, the clay particles acquire a negative charge due to interactions with the brine. The existence of negative charges induces repulsive forces throughout the formation process, which subsequently induces the movement of clay particles after detachment. The movement of clay particles during migration might damage pore throats, diverting and altering the flow of injected brine down alternative pathways. This subsequently facilitates the mobilisation of residual oil from regions that have not been effectively swept. Furthermore, the fines migration may either enhance or hinder oil recovery by reducing permeability (Agi et al., 2024). Fine migration in low-permeability areas can cause high-permeability formation damage, redirecting water from low to high-permeability zones and resulting in early water breakthroughs and decreased macroscopic efficiency (Dordzie and Dejam, 2021). The fines migration in highly permeable areas can cause pore plugging. Consequently, increasing tortuosity during LSWF in these areas may cause water to slow and change direction within the swept zone. This delays the breakthrough and redirection to unswept areas of the reservoir, ultimately leading to improved oil recovery, as shown in Fig. 1.

Bernard (1967) established a correlation between clay swelling, fines migration, and enhanced recovery. According to his hypothesis, clay disintegrates into exceedingly fine particles upon contact with water. These particles traverse the well-defined flow paths and ultimately obstruct them. Following the complete or partial closure of the existing flow channels, new flow channels form. This results in enhanced oil production when these new channels get saturated. In a study by Smith (1942) a series of laboratory experiments was conducted on sandstone cores using both fresh water and a calcium chloride solution. Oil recovery was greater with a brine solution than with fresh water. This difference may be attributed to the phenomenon of clay swelling that occurs when fresh water is used. The phenomenon of clay swelling and dispersion, followed by an elevation in differential pressure, was originally observed by Martin (1959) and subsequently confirmed

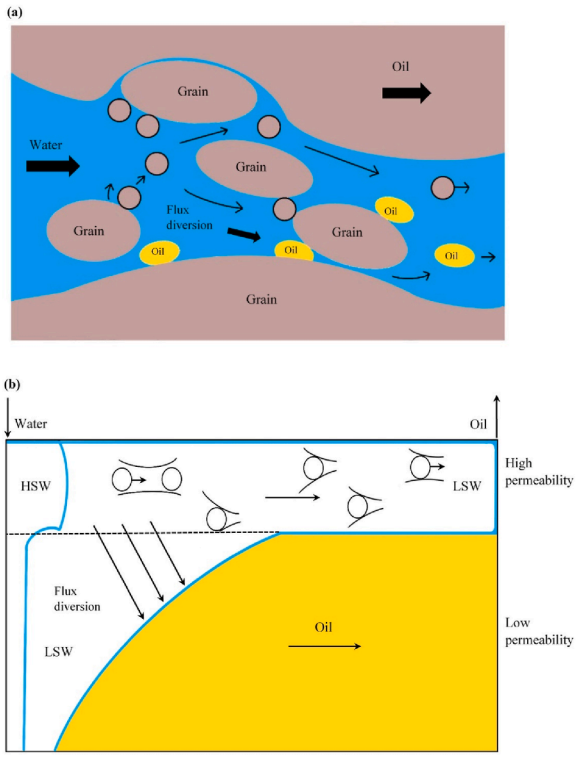


Fig. 1. A schematic representation of (a) fines migration and flux diversion due to formation damage, (b) increased oil recovery displacement caused by fines migration (Chequer et al., 2019; Dordzie and Dejam, 2021).

by Bernard (1967). The study conducted by Martin (1959) focused on the use of LSWF for improved oil recovery on sandstone cores. The author observed a positive correlation between the reduction in salinity of the injected brine and the subsequent rise in oil recovery. The author hypothesised that this increase might be attributed to processes related to emulsification and clay swelling. Significantly, this observation was further associated with improved oil recovery, underscoring the crucial role of clay behaviour in the broader dynamics of oil extraction.

According to another finding by Kia et al. (1987), exposure of sandstones to sodium salt solutions before flooding with fresh water led to the release of clay particles and a significant reduction in permeability. Tang and Morrow (1999) observed a decrease in brine permeability due to the movement of fines. The presence of particles with potential mobility, such as kaolinite, has a substantial effect on improving oil recovery. Furthermore, the combination of fine migration, initial water saturation, and crude oil adsorption is essential for achieving improved oil recovery while reducing salinity levels. In their experimental study, Al-Sarihi et al. (2018) examined the influence of fines migration on the displacement of residual oil during LSWF. The authors documented that the migration of fines can substantially influence residual oil saturation. They observed a significant decline in residual oil saturation due to microscale flow diversion induced by migrating fines. In another study, Tangparitkul et al. (2020) examined fine detachment and migration by modifying critical retention concentration in relation to stress and salinity dependencies. The authors observed that particle dissociation and clay swelling during LSWF lead to a decrease in the effective flow channel and pore space. Clay swelling depends on the size of the original clay particle, and diminutive particles may remain adhered to the surface due to insufficient detachment force.

Despite the favourable outcomes reported in the research, this mechanism is inconsistent with other investigations (Lager et al., 2008; Morrow and Buckley, 2011; Rezaeidoust et al., 2009; Zhang et al., 2007b; Zhang and Morrow, 2006). According to Lager et al. (2008), no significant movement of fines migration was observed in LSWF experiments conducted for improved oil recovery under various conditions. Zhang et al. (2007b) observed the absence of clay content during the production stream and the oil/brine interaction. Furthermore, there was no evidence of pore-throat obstruction due to flow diversion. Based on the research by Berg et al. (2010), the experimental observations throughout the process of increased oil recovery did not indicate any fines migration. This suggests that the improved recovery achieved by LSWF cannot be ascribed to fines migration. Consequently, it can be concluded that no observable correlation exists between clay migration and improved oil recovery with LSWF. Conversely, the inclusion of clay has the potential to induce a modification in wettability. This occurs by liberating oil-wet particles from the rock surface, thereby transforming the system into a more water-wet state (Liu and Wang, 2020). The role of fine migration in the context of LSWF is seen as supplementary rather than primary.

2.2. Effect of pH

McGuire et al. (2005) postulated a mechanism that integrates the effect of pH on the LSWF. The mechanisms responsible for LSWF may be attributed to elevated pH and decreased IFT, which resemble the processes observed in alkaline flooding (McGuire et al., 2005). The technique includes cation exchange between the clay particles and the injected brine. The injection of LSW can raise pH, reducing IFT between the brine and oil and thereby enhancing the capillary number and improving oil recovery. The observed rise in pH may be attributed to hydrogen-ion exchange between water molecules and adsorbed sodium ions, as shown by Mohan et al. (1993). Higher pH levels lead to a chemical transformation of the organic acids in crude oil. The organic acids present in crude oil undergo a reaction that forms surfactants (specifically, soaps) within the oil itself. These surfactants have the ability to reduce the IFT between oil and water. The soap produced assists in forming oil/water or water/oil emulsions due to its reduced IFT. The process of emulsification has the potential to enhance the efficiency of water sweep (Sheng, 2011). The clay exhibits unique phenomena of cation exchangers due to its substantial surface area. During the process, clay undergoes adsorption of both organic and inorganic substances from the surrounding formation water (Austad, 2013). This adsorption process is particularly pronounced for calcium ions (Ca^{2+}), leading to the establishment of a chemically stable equilibrium within the reservoir, influenced by factors such as pressure, temperature, and pH. The infusion of brine into the formation disrupts the equilibrium between the clay surface and brine, leading to the desorption of Ca^{2+} cations.

Following the adsorption of the H^+ proton onto the clay surface, which replaces the previously occupied Ca^{2+}, a subsequent rise in pH occurs on the clay surface, as demonstrated in Eq. (1). After the rise in the pH levels in the vicinity of the clay surface, chemical interactions take place between the naturally occurring acidic and adsorbed basic substances, as explained in Eqs. (2) and (3) (Austad et al., 2010). Therefore, as a result of these processes, the clay surface undergoes a transition from a water-wet to a mixed-wet state. Consequently, crude oil desorption from the clay surface is enhanced, leading to a greater amount of recoverable oil (Katende and Sagala, 2019; Vledder et al., 2010).

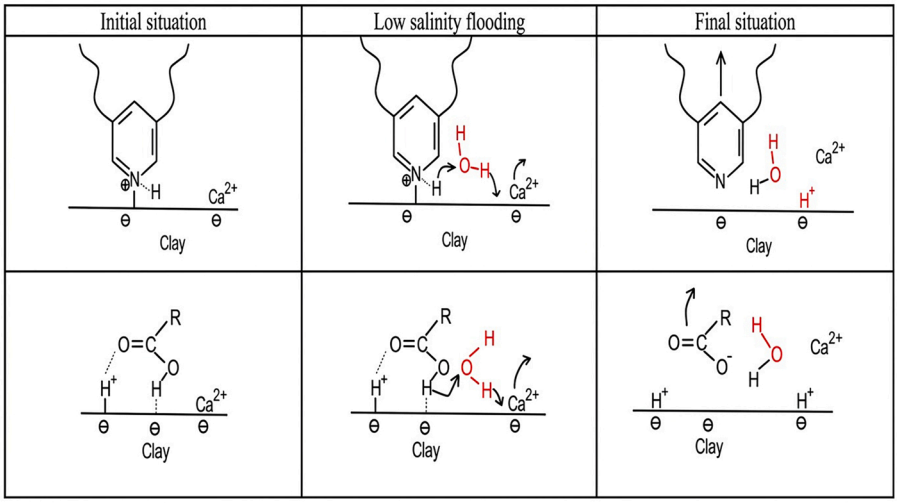


Fig. 2. Effect of pH increases on low-salinity waterflooding mechanism. Desorption of basic and acidic materials at the upper and lower parts (Austad et al., 2010).

$$Clay - Ca^{2+} + H_2O \leftrightarrow Clay + H^+ + Ca^{2+} + OH^- \quad (1)$$

$$Clay - NHR_3^+ + OH^- \leftrightarrow Clay + R_3N + H_2O \quad (2)$$

$$Clay - RCOOH + OH^- \leftrightarrow Clay + RCOO^- + H_2O \quad (3)$$

In carbonate reservoirs, the pH of the formation water typically ranges from 7 to 9, indicating a slightly alkaline nature. The introduction of diluted seawater may raise pH levels due to calcite decomposition (Austad et al., 2005; Derkani et al., 2018). The dissolution of carbonate minerals is due to acid-base interactions at the pore or rock surface. This results in an increase in the pH of the brine, which subsequently changes the reservoir from oil-wet to mixed or water-wet. Consequently, the oil can flow freely towards the production well without any obstruction. Moreover, the observed pH elevation might trigger the activation of inherent surfactants, leading to reduced IFT and the formation of emulsions. These emulsions are essential for enhancing the recovery of residual oil saturation in acidic oil reservoirs (Awolayo et al., 2018; Bhicajee and Romero-Zerón, 2021; Yousef et al., 2011). The impact of pH increases on the LSWF mechanism is shown in Fig. 2.

2.3. Multicomponent Ion exchange

Lager et al. (2008) introduced the multicomponent ion exchange (MIE) mechanism. The MIE process entails the desorption of organic compounds, including positively and negatively charged, from the surface of the clay particles, as shown in Fig. 3. This desorption process is initiated by divalent cations, particularly calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions, which originate from the injected brine. Cationic chemicals present in crude oil, such as nitrogen (N^+), undergo adsorption onto the negatively charged clay surface via cation exchange. While divalent cations, namely Ca^{2+} and Mg^{2+}, bind the crude oil's negatively charged acidic components and the negatively charged minerals on the surface of the clay. This bridging is facilitated by cation, ligand, and water interactions. Seccombe et al. (2010) demonstrated the occurrence of MIE effects from the perspective of LSWF. Their findings revealed a positive relationship between reduced salt content in injected brine and increased oil production rate.

Ion exchange at the rock-brine and crude oil-brine interfaces detaches crude oil from the rock surface, thereby altering wettability to a more water-wet state. However, to facilitate MIE processes during LSWF, the crude oil must contain polar components, the rock surfaces must contain clay minerals, and the connate water must contain divalent cations. The adsorption processes of the organic compound in relation to its impacts on MIE are presented in Table 1. The examination of the adsorption of pore water components in the marine environment onto the montmorillonite clay mineral has provided insights into the magnitude and fundamental mechanisms involved. Among the eight mechanisms postulated by Schroth and Sposito (1997), van der Waals interactions, ligand exchange, and cation bridging emerge as the primary processes governing this adsorption phenomenon. According to the DLVO theory, elevated ionic strength results in weaker electrostatic repulsion forces. Consequently, the particles may come into proximity owing to the significant influence of van der Waals attractive forces at elevated ionic strengths.

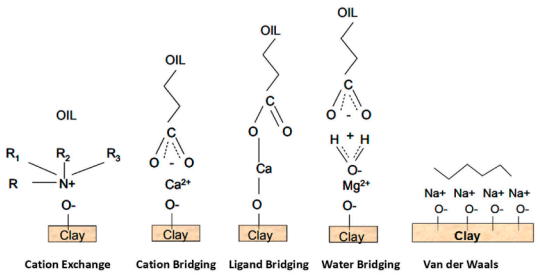


Fig. 3. Diverse adhesion mechanisms between crude oil (polar components) and clay surface (Lager et al., 2008).

Table 1
Mechanism of organic compound adsorption onto minerals for MIE effects (Schroth and Sposito, 1997).

Mechanisms	Organic functional group involved
Cation exchange	Amino, ring NH, heterocyclic N (aromatic ring)
Protonation	Amino, heterocyclic N, carbonyl, carboxylate
Anion exchange	Carboxylate
Water bridging	Amino, carboxylate, carbonyl, alcoholic OH
Cation bridging	Carboxylate, amines, carbonyl, alcoholic OH
Ligand exchange	Carboxylate
Hydrogen bonding	Amino, carbonyl, carboxyl, phenolic OH
Van der waals interactions	Uncharged organic units

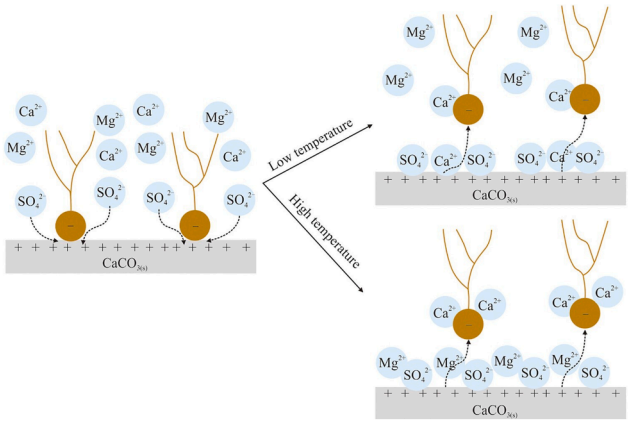


Fig. 4. Mechanism of multiple ion exchange in carbonates during low-salinity waterflooding (Awolayo et al., 2018; Zhang et al., 2007a).

Furthermore, it has been suggested that MIE may alter the wettability of sandstones and carbonates. In sandstone, ionic exchange occurs due to the presence of clay minerals. This exchange occurs between the cations in the low-salinity water and the negatively charged clay particles. Subsequently, the oil adsorbed on the clay mineral surfaces is released, enhancing the surface's water-wettability (Jackson et al., 2016). Whereas, in carbonates, the addition of sulphate ions $\left(SO_4^{2-}\right)$ in conjunction with the injected brine, induces the migration of Ca^{2+} from the interior of the carbonate core sample to its surface, where they form robust intermolecular bonds with one another, as shown in Fig. 4. Consequently, the oil that was previously adhered to the carbonate rock surface becomes dislodged as a result of ion exchange, leading to a shift in the wetting characteristics of the rock towards a lesser affinity for oil (Dordzie and Dejam, 2021). The equations for MIE during LSWF are presented as Eqs. (4) and (5) for sandstones (Ridwan et al., 2020) and Eq. (6) for carbonates (Yutkin et al., 2022).

$$>Si-O-M-COO-R<+2Na^{+}\leftrightarrow>Si-ONa+Mg^{2+}+NaCOOR< \quad (4)$$

$$>Al-O-M-COO-R<+2Na^{+}\leftrightarrow>Al-ONa+Mg^{2+}+NaCOOR< \quad (5)$$

$$Rock-Ion-Adsorbed\ oil+Ions\leftrightarrow Rock-Ion+Release\ Ion+Ion-Desorbed\ Oil$$

$$2>S^{-}Na^{+}+Ca^{2+}\overset{K_{CaNa}}{\leftrightarrow}>S_2^{2-}Ca^{2+}+2Na^{+} \quad (6)$$

where $2>S^{-}$ represents the surface exchange site, while K_{CaNa} is the exchange equilibrium constant.

Zaeri et al. (2018) conducted LSWF studies in carbonates using spontaneous imbibition and contact angle measurements. They observed that the primary mechanism responsible for the change in wettability and the improvement in oil recovery throughout the investigation was the MIE. In another research study by Bhicajee and Romero-Zerón (2021), the authors examined the operational principles of LSWF to improve heavy oil recovery in unconsolidated sandstone formations. The authors suggest that the transition from mixed-wet to water-wet wettability may be responsible for the observed increase in hydrocarbon recovery. This observed wettability reversal was attributed to MIE. Similarly, Zhang et al. (2007a) investigated the interaction among the potential-determining ions (PDI) (Ca^{2+}, Mg^{2+}, and $SO4^{2-}$) through spontaneous imbibition on a chalk surface. The chalk surface exhibited a rise in the degree of Mg substitution for Ca^{2+}, which subsequently resulted in the desorption of the carboxylic acid at elevated temperatures. This phenomenon ultimately enhances the water-wetness of the chalk surface.

2.4. Mineral dissolution

The mechanism of mineral dissolution was suggested by Pu et al. (2010). The process was derived from an experimental study conducted on sandstone containing anhydrite cement and clay within the Wyoming Formation. The absence of a high clay content precluded the occurrence of these alternative mechanisms. Consequently, following the core flooding experiment utilising a brine solution containing sodium chloride (NaCl), the effluent is found to contain sodium ions (Na^{+}), magnesium ions (Mg^{2+}), and potassium ions (K^{+}). The observed elevated effluent levels provide empirical support for the mineral dissolution mechanism originating from anhydrite cement. This phenomenon has notable implications for alterations in wettability and subsequent reductions in permeability subsequent to the liberation of dolomite crystals. The oil recovery rate improved significantly, rising from 5% to 8% of the original oil in place (OOIP) across all cores tested.

The process of mineral dissolution can substantially improve oil recovery by altering wettability, as shown in Fig. 5. This modification of wettability has been attributed to the dissolution of calcite and anhydrite (Hiorth et al., 2008; den Ouden et al., 2015; Chandrasekhar et al., 2018; Yousef et al., 2011; Austad et al., 2015). Calcium carbonate ($CaCO_3$) dissolves in brine. This dissolution of calcium carbonate minerals in brine is contingent upon the concentration of the solution (den Ouden et al., 2015). The dissolution reaction can either enhance or degrade the petrophysical characteristics of rocks, such as porosity and permeability (Isah et al., 2022). The dissolution process typically results in an increase in the pH of brine (Alotaibi et al., 2018). The dissolution rate is influenced by several factors, including the solution's ionic strength, the surface area of the dispersed particles, and the solution's temperature (Finneran and Morse, 2009; Morse and Arvidson, 2002; Morse et al., 1979).

According to Austad et al. (2015), the LSWF in limestone cores leads to an elevation in sulphate ion (SO_4^{2-}) levels owing to the dissolution of anhydrite and reduction in NaCl concentration in the injected brine. The modification of NaCl and SO_4^{2-} concentration can alter the wettability of the core to a water-wet state. Alagic et al. (2011) examined the effect of LSWF on both the aged and unaged cores of Berea sandstone. Following LSWF, the effluent concentration investigation showed that although Ca^{2+} ions were discharged from both cores, Mg^{2+} ions were retained firmly in the cores. The mineral breakdown process that occurs during LSWF was proposed as the cause of Ca^{2+} ion discharge. Furthermore, pressure drops with irregular spikes, and reductions were

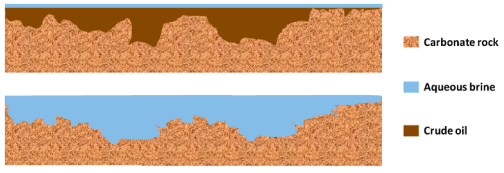


Fig. 5. Wettability alteration due to mineral dissolution (a) before dissolution (b) after dissolution (Sagbana et al., 2023).

observed throughout the unaged cores, even after production ended. In aged cores, this very erratic pressure profile was not observed. The significant swelling of the clays was likely to result in a noticeable, steadily growing pressure buildup throughout the core. The introduction of fines into the fluid flow and their movement towards pore constraints, leading to potential blockages, may result in fluctuating pressure changes (both increase and decrease) and a gradual rise in pressure drop. This phenomenon was evident in LSWF conducted on unaged cores, but not on aged cores. Even after oil production stopped completely and only the water phase was produced, pressure increased consistently throughout the cores. Mineral dissolution or fines release that hinders fluid flow might explain the uneven pressure profiles seen in unaged cores, as well as the extended oil recovery in unaged cores. McGuire et al. (2005) reported an increase in hydrocarbon recovery of 6%–12% subsequent to performing numerous LSWFs in sandstones. This was attributed to the system's pH rising to 9.

Pu et al. (2010) introduced the method after conducting an experiment involving the cores of a phosphoria dolomitic reservoir. Following core flooding experiments with brine containing NaCl, the effluent showed a reduction in permeability, as corroborated by Lebedeva et al. (2009). The presence of a significant volume of effluent indicates dolomite mineral dissolution and suggests alterations in wettability. The dissolution process results in a significant improvement in oil recovery through wettability alteration. Kafili Kasmaei and Rao (2015) observed a significant alteration in the contact angle of dolomite cores due to the application of LSWF. Specifically, they observed a transition from a very oil-wet condition, characterised by a contact angle of 158°, to an intermediate wet state with a contact angle of 113°. The potential mechanisms for altered wettability could include mineral dissolution and fines migration.

Another study by Al-Bayati et al. (2022) examined the effect of LSWF in carbonate reservoirs and explored the key factors influencing oil recovery improvement. The trials were carried out under both steady and unstable conditions. In comparison to HSWF with 0.6 M NaCl, a significant change in the relative permeability curve was observed when LSWF with 0.2 M NaCl was employed under steady-state conditions. The initial water saturation and relative permeability decreased throughout the drainage process. The relative permeability curves for oil and brine shifted to the right throughout the imbibition process, indicating an increase in water-wetness. Furthermore, calcite dissolution altered the petrophysical properties of the carbonate cores, including permeability, porosity, and pore volume. den Ouden et al. (2015) reported that there was a lack of significant alteration in the porosity of the rock samples, as well as an absence of detectable modifications in the pore structures, following the injection of a 2000 ppm NaCl brine for a duration of 53 pore volumes. This assessment was made using micro-CT imaging of the flooded core. The researchers concluded that, even if calcite dissolution occurs, it may not affect wettability alteration or improve oil recovery.

2.5. Electrical double layer expansion

Winsauer and McCardell (1953) were the first to propose the notion of ionic double-layer conductivity in reservoirs. Their findings demonstrated that the specific ions and their concentration in the electrolyte influence the magnitude of additional double-layer conductivity. The electrical double layer (EDL) describes the formation of two charged layers on a charged colloidal particle's surface when dispersed in an aqueous solution. The Stern layer is a stable, firmly attached layer composed of counterions derived from water that neutralise the colloid's charge. The diffuse

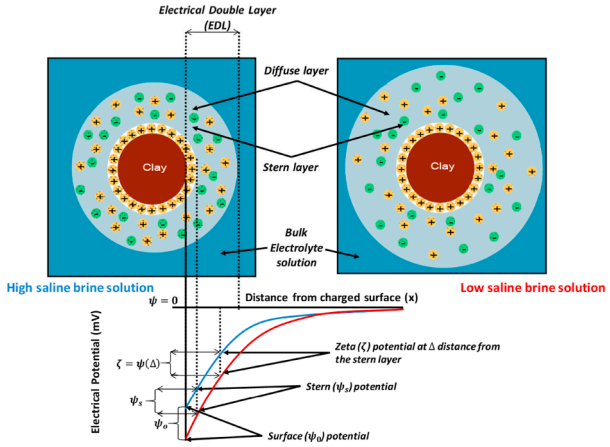


Fig. 6. Electrical double layer around a charged particle and zeta potential (Purswani et al., 2017).

layer forms from weakly bonded ions in the outer region. The zeta potential indicates the thickness of the EDL, as shown in Fig. 6.

According to Knott (2009), when a negatively charged clay particle interacts with water in the porous rock matrix of an oil-bearing reservoir, an EDL will establish over the particle. The dual layers consist of an inner layer of adsorbed positive ions and an outer layer mainly composed of scattered negative ions. The reduction in brine salinity in the system generates an opposing force, resulting in the expansion of the EDL at the interfaces between the rock and brine and between the crude oil and brine. This results in an increase in the water film thickness at the interface, leading to a change in wettability towards greater water wetness (Brady et al., 2015).

Ligthelm et al. (2009) suggest that the fundamental mechanism responsible for improving oil recovery during LSWF is wettability alteration towards a water-wet state. This modification results from the growth of the EDL at the interfaces between minerals and brine. They observed that the injection of low-salinity brine with reduced multivalent cations in the presence of negatively charged

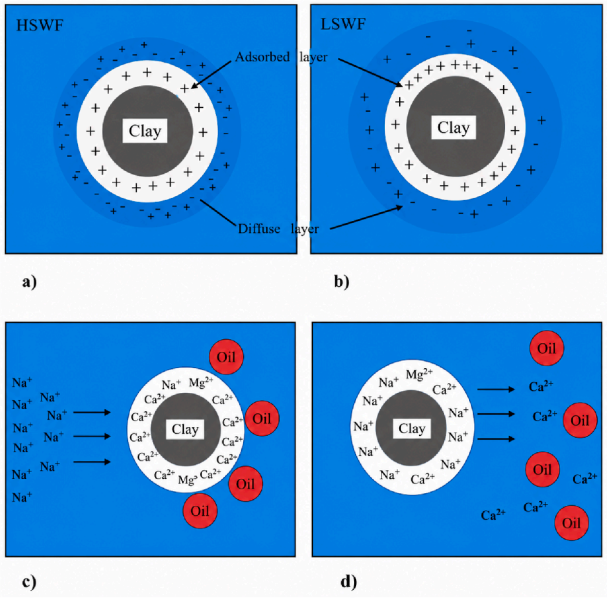


Fig. 7. Effect of low-salinity waterflooding on electrical double layer expansion (Mwakipunda et al., 2023).

clay surfaces results in the development and expansion (diffusion) of the EDL into the oil phase, as evidenced by the zeta potential. This generates a repulsive force that exceeds the adhesive force between the oil and the clay surface, resulting in a change in wettability from oil-wet to mixed-wet or water-wet, as shown in Fig. 7. The magnitude of the double layers is contingent upon the quantity of ions present in the aqueous solution. The compactness of the double layer is shown to increase in environments of high salinity, characterised by a higher concentration of ions in the water (Fig. 7a). Conversely, introducing low-salinity water results in an enlargement of the double layer, as depicted in Fig. 7b. The layer of adsorbed positively charged ions (Ca^{2+} or Mg^{2+}) serves as a connector between the clay and oil droplets. The low-salinity brine facilitates the expansion of the diffuse layer, therefore allowing the entry of monovalent ions (Na^{+}) into the double layer from the injected brine, as illustrated in Fig. 7c. The electrostatic repulsion between clay particles and oil is enhanced by the monovalent ions, which displace the divalent ions. The connections between oil and clay particles are disrupted when repulsive forces exceed binding forces due to multivalent cation bridges, leading to the desorption of oil particles from the clay surfaces. The wetting condition is altered by the decrease in the rock surface area covered with crude oil, facilitating the displacement of oil from the reservoir, as seen in Fig. 7d.

Sari et al. (2019) assessed the potential of LSWF by integrating contact angle, zeta potential, and the surface complexation model. The findings validate that EDL caused the alteration in carbonate wettability. Furthermore, they suggest that the reduction in the brine salinity resulted in higher negative electrostatic charges at the brine/calcite and oil/brine interfaces. Hilner et al. (2015) documented a substantial reduction in adhesion between quartz surfaces and non-polar oil molecules at a salinity threshold comparable to that typically associated with the low-salinity effect. They proposed that DLE would consistently influence the LSWF, regardless of the involvement of other processes. Fathi et al. (2010) investigated the impact of salinity and ion composition on chalk wettability modification using chromatographic and spontaneous imbibition wettability experiments. They concluded that the EDL effect is essential for wettability modification in chalk. Heberling et al. (2011) The authors examined the zeta potential of calcite in LSWF using a Stern surface complexation model. They suggested that the dynamics of the ELD may largely explain the alteration in calcite wettability at the interface. In contrast, Kazankapov (2014) investigated LSWF in carbonates through spontaneous imbibition and chromatography wettability experiments. In their investigation, they determined that the EDL did not contribute to the alteration in the wettability of the carbonates.

2.6. Effect of salt-in

The salt-in mechanism, postulated by Rezaeidoust et al. (2009), requires lowering the salt concentration inside the system. Adding salt to a solution can significantly decrease the solubility of organic compounds in water, a phenomenon known as the salting-out effect. Whereas, the removal of salt from water may increase solubility, referred to as the salting-in effect (Rezaeidoust et al., 2009). The introduction of LSW into the reservoir disrupts the balance between water, oil, and rocks. This disruption results in enhanced solubility of polar organic substances in water. Some inorganic substances present in the injected water facilitate the dissociation of water molecules, thereby enhancing the solubility of organic substances. Furthermore, the solubility of organic material may increase when salinity decreases below a certain ionic strength, thereby enhancing oil recovery. This resembles the transition from oil-wet to more water-wet conditions. The enhanced solubility of organic material in the brine can be likened to the desorption of oil droplets from the clay mineral surfaces. This process was confirmed in a preliminary investigation by Rezaeidoust et al. (2009). They observed that the desorption of 4-tert-butyl benzoic acid in brine, including kaolinite, began to rise with a reduction in the salinity of the brine. However, this process does not account for the correlation among mineral composition, pH, salinity, and other factors. Furthermore, it has also received little attention and has not been well explored compared to other mechanisms (Rezaeidoust et al., 2011).

2.7. Chemical osmosis

The concept of chemical osmosis as an LSWF process for improving oil recovery was first presented and described by Sandengen and Arntzen (2013). Conventional LSWF methods primarily aim to recover oil by injecting low-salinity water. Conversely, chemical osmosis is regarded as a mechanism for extracting crude oil from hydraulically stagnant, unreachable pores using direct LSWF methods. The effectiveness of this mechanism depends on the presence of high-salinity connate water. Additionally, the system must exhibit an oil-wet characteristic to facilitate the separation of connate water by forming oil films (Liu and Wang, 2020).

Chemical osmosis is the process by which water molecules migrate across a semipermeable membrane, driven by the difference in chemical potential of water on either side. Crude oil functions as a semi-permeable barrier, selectively permitting the diffusion of pure water injected to move towards the connate water. The entry of water molecules from the low-salinity water increases the pressure and volume of the high-salinity water. The rise in pressure and volume prompts water migration within the hydraulically stagnant pores towards the injected low-salinity water. This expansion process thus improves oil recovery, as shown in Fig. 8 (Fang et al., 2020; Takeda et al., 2019; Yan et al.,

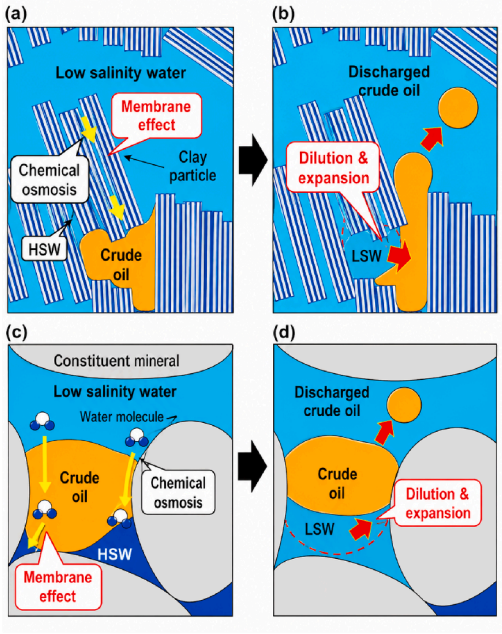


Fig. 8. Chemical osmosis-driven oil extraction through semipermeable membranes in dead-end pores (a) Chemical osmosis through interlayer spaces in clay minerals (b) oil extraction due to the dilution and expansion of high-salinity connate water (c) Chemical osmosis through crude oil and water film adsorbed on the hydrophilic mineral surface and (d) oil extraction due to the dilution and expansion of high-salinity connate water (Takeda et al., 2021).

2020). Furthermore, several studies suggest that the rock matrix may function as a membrane; however, this may depend on the rock type (Kurtoglu et al., 2014).

The results from many investigations conducted by various researchers provide substantial information on osmotic water transport and oil mobilisation across different wettability conditions (Callegaro et al., 2013; Fredriksen et al., 2016, 2017; Pollen and Berg, 2018; Rotondi et al., 2014; Sandengen et al., 2016; Yousef and Ayirala, 2014). However, the extent to which osmotic events contribute to improved recovery remains poorly supported by empirical evidence. In addition to the constraints of requiring oil-wet systems and high-salinity water, the osmosis process does not account for the observed sensitivity to ion type observed in some LSWF studies. Therefore, it is plausible that this mechanism does not significantly contribute to the LSWF process (Liu and Wang, 2020).

2.8. Interfacial tension reduction

McGuire et al. (2005) first proposed reducing interfacial tension (IFT) at the crude oil-aqueous solution interface as an effective technique for improving oil recovery in LSWF. The implementation of LSWF can lead to a reduction in IFT between crude oil and brine. Furthermore, the breakdown of polar components in oil tends to produce surfactants in the reservoir and functions as an alkaline flooding technique. This surfactant decreases IFT between water and oil and increases pH, thereby enhancing water wettability and improving oil recovery. Mokhtari and Ayatollahi (2019) examined the decomposition of crude oil's polar components under LSWF conditions. They suggest a specific ionic strength at which the decrease in IFT is maximised. The interaction of naphthenic acid in crude oil with brine leads to a reduction in the pH of the brine. This process may also alter the system's wettability, rendering it more water-wettable. The influence of fluid-fluid interaction during LSWF was investigated by Mokhtari et al. (2019). The researchers observed that the polar constituents of the crude oil dissociate as the brine salinity decreases. This results in alterations to the brine's interfacial and physical properties. The brine exhibited the lowest IFT value when diluted with saltwater at a ratio of two parts seawater to one part brine. Furthermore, the capillary number is effectively reduced by alterations in wettability and by lowering IFT. Subsequently, these modifications have a discernible impact on residual oil recovery. On the contrary, several studies have expressed the viewpoint that LSWF has little or negligible influence on the process of reducing IFT. Consequently, the mechanism cannot be categorised as the primary process for improving oil recovery during LSWF (Al-Harrasi et al., 2012; Lager et al., 2008; Yousef et al., 2010, 2011).

2.9. Formation of fluid microdispersion

Microdispersion refers to the presence of small pockets of brines that are surrounded by polar components of crude oil. Direct flow visualisation may be used to investigate the occurrence of brine microdispersion in the oil phase near the oil-brine interface during LSWF. While the impact of LSWF on lowering IFT is not substantial, it does lead to the diffusion of surface-active materials (SAMs) at the fluid interface, such as asphaltenes and naphthenic acids. This diffusion process results in the formation of a microdispersion within the crude oil phase (Mehraban et al., 2021). The significance of microdispersion lies in its capacity to induce subsequent changes in the wettability of the rock surface (Emadi and Sohrabi, 2013). Moreover, the phenomenon of partitioning SAM might be attributed to the formation of solvation shells around them, as discussed by Shahmohammadi et al. (2022).

The formation of microdispersion is substantial, as it leads to a subsequent shift in the rock surface's wettability. The polar constituents present at the oil-brine interface in crude oil are drawn towards the dispersed components within the bulk oil. The charging characteristics of the oil-brine interface are altered, resulting in the disruption of bonds formed by these polar constituents. Consequently, the system exhibits an increased affinity for water, owing to intensified EDL repulsion and diminished bonding between the crude oil and the rock. Furthermore, the formation of microdispersion may directly influence multiphase flow, similar to the osmosis process (Sohrabi et al., 2017). However, this phenomenon is regarded as a distinct instance of osmosis rather than being classified as an alternative LSWF process.

The study conducted by Emadi and Sohrabi (2013) employed micromodel testing to investigate the influence of fluid-fluid interaction on the LSWF process. The results suggest that injecting low-salinity water into reservoirs leads to the formation of larger water microdispersion particles and partitioned surface-active materials (SAM). This occurs when low-alinity water interacts with crude oil, either at the interface or within the crude oil itself. The denser nature of these dark particles facilitates their downward movement towards the clay surface, resulting in increased water wettability by removing SAM. Consequently, this alteration in wettability results in an improved oil recovery, as shown in Fig. 9.

Additionally, Fattahi Mehraban et al. (2020) analysed the water microdispersion generated during LSWF using Fourier Transform Infrared (FTIR) spectroscopy. The authors observed that the acidic constituents of crude oil facilitated the development of water microdispersions. These microdispersions significantly influence surface wettability and induce oil swelling, thereby improving oil recovery. Research by Darvish Sarvestani et al. (2019) revealed that the contact length does not influence the production of microdispersion. The level of dilution of injected brine is contingent upon many factors. Moreover, seawater diluted 10 times showed higher quantities and larger emulsion droplets after contact with crude oil for 0 and 10 h, compared to undiluted seawater. Maaref et al. (2017) have shown comparable findings at the micromodel level. AlHammadi et al. (2018) examined the influence of microdispersion, a fundamental mechanism for improving oil recovery

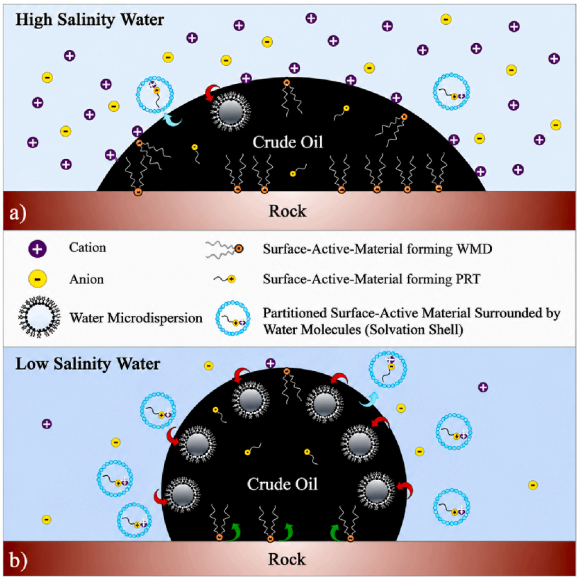


Fig. 9. Microdispersion and partitioning (a) high-salinity water, and (b) low-salinity water (Mehraban et al., 2021).

during LSWF, in carbonate cores. A selection of three crude oils was made from a larger pool of screened oils to conduct core-flooding studies. The crude oils, denoted A and B, exhibit identical characteristics but differ in their respective tendencies to create microdispersions. Crude oil A has a strong propensity for microdispersion formation, while crude oil B shows a weak tendency. Conversely, the final crude oil sample, denoted C, is derived from crude oil A by eliminating surface-active components responsible for the formation of microdispersions. Following the completion of core-flooding studies, it was observed that crude oil A exhibited a notable incremental oil recovery of 6.3%. Furthermore, this sample shows a pronounced tendency toward microdispersion formation. In contrast, flooding studies conducted with crude oil B showed a low tendency to form microdispersions. Crude oil C, which was treated to remove the surface-active components, did not result in any significant increase in oil recovery. Consistent findings were observed in the following studies of LSWF in sandstone cores with high clay content (Fattahi Mehraban et al., 2020).

2.10. Viscoelasticity

The viscoelasticity of the oil-water interface is another process associated with liquid-liquid interactions (Alvarado et al., 2014). The viscoelasticity of the oil-water interface is enhanced by a decrease in ionic strength, irrespective of the specific kinds of ions present. This heightened interfacial viscoelasticity contributes to oil recovery via two distinct mechanisms. Initially, snap-off in water-wet cores is mitigated, reducing the amount of oil trapped during low-salinity waterflooding. Moreover, it has been shown that the size of trapped oil droplets tends to increase during snap-off events, potentially leading to the obstruction of pore throats and the creation of new channels for fluid flow. A summary of LSWF mechanisms is presented in Table 2.

3. Discrepancies in low-salinity waterflooding mechanisms

The alteration in wettability remains the primary outcome of most proposed LSWF mechanisms. However, the driving physicochemical processes differ substantially depending on lithology, brine chemistry, and crude oil composition. Although numerous mechanisms have been proposed to explain the effectiveness of LSWF, the literature presents several apparently conflicting findings, particularly regarding fines migration, IFT reduction, and the relative dominance of MIE. These inconsistencies do not necessarily invalidate the proposed mechanisms; rather, they reflect the strong dependency of LSWF behaviour on reservoir-specific parameters and experimental conditions. A critical synthesis suggests that variations in crude oil composition, rock mineralogy, aging protocols, salinity window, temperature, and scale of investigation collectively contribute to the divergent observations reported in previous studies.

- Crude oil composition plays a critical role in governing fluid–rock interactions and overall displacement efficiency during recovery processes. The concentration of polar fractions, such as resins and asphaltenes, significantly influences interfacial tension behaviour, wettability alteration, and adsorption characteristics at the rock surface. Crude oils with higher TAN or greater polarity tend to exhibit stronger electrostatic interactions at the COBR interface, thereby enhancing the potential for wettability modification under low-salinity conditions. Conversely, low-polarity oils may show limited responsiveness, leading to reports that question the role of ion exchange or IFT reduction. Thus, discrepancies in crude oil composition across experimental studies can substantially affect mechanistic interpretation.
- Mineralogy plays a decisive role in governing fines migration and surface charge behaviour. Sandstones with reactive clay minerals (e.g., kaolinite, illite, smectite) are more prone to fines detachment under reduced salinity conditions, while clean quartz-rich sandstones may show minimal response. Similarly, in carbonates, the role of sulphate ions and mineral dissolution is more pronounced. Variations in clay type, distribution, and cation exchange capacity (CEC) contribute significantly to the divergent observations reported in fines migration studies.
- Laboratory aging procedures strongly influence experimental outcomes by controlling wettability alteration and fluid–rock interactions. The parameters, such as aging time, temperature, and crude oil exposure, govern the development of wettability states and the adsorption of polar oil components onto mineral surfaces. Insufficient aging may produce artificially water-wet systems, thereby masking potential LSWF effects. Therefore, differences in aging protocols across studies explain discrepancies in reported wettability alterations and improvements in recovery.
- LSWF effectiveness is not governed solely by total salinity reduction but by the existence of an optimal “salinity window.” Excessive dilution may suppress beneficial ionic interactions, whereas insufficient dilution may fail to trigger EDL expansion or MIE processes. Moreover, the type and concentration of PDIs (Ca^{2+}, Mg^{2+}, SO_4^{2-}) significantly influence the mechanism. Studies employing different dilution factors or brine chemistries may therefore report opposing results regarding IFT reduction or wettability shifts.
- Temperature affects ion exchange kinetics, mineral dissolution rates, and crude oil polarity. At elevated temperatures, certain mechanisms such as mineral dissolution may become more significant, whereas fines migration may be less dominant. Thus, variations in experimental temperature and pressure conditions across studies contribute to inconsistencies in mechanistic interpretation.
- Finally, many contradictory findings stem from scale-dependent effects. Core-scale experiments may exaggerate fines migration or localised wettability shifts, while field-scale responses integrate heterogeneity, geochemical reactions, and reservoir connectivity. The limited number of full-field LSWF implementations further complicates direct validation of laboratory mechanisms.

4. Machine learning

Machine learning (ML), a subset of artificial intelligence (AI), has recently gained significant popularity due to its widespread use in simulating complex problems across multiple domains. ML employs mathematical models to perform regression, classification, and clustering. ML employs computer algorithms to perform these tasks autonomously, leveraging acquired expertise and data (Mitchell, 1997). ML relies on acquiring knowledge of patterns and making inferences from data to predict. ML methodologies may be broadly classified into three categories: supervised learning (including regression and classification), unsupervised learning (including clustering), and reinforcement learning (including decision-making), as shown in Fig. 10.

4.1. Machine learning algorithms in low-salinity waterflooding

ML has undergone almost half a century of development, resulting in a multitude of algorithms. The application of ML

Table 2
Summary of low-salinity waterflooding mechanisms.

Mechanism	Dominant rock type	Required conditions	Primary recovery Driver	Strength of experimental evidence	Field validation status	Key limitations
Fines migration	Sandstone (clay-rich; especially kaolinite-bearing)	Presence of migratable clays; significant salinity reduction; initial water saturation; crude oil adsorption	Flow diversion, pore plugging, and wettability alteration	Moderate but inconsistent; several lab confirmations, but also contradictory studies	Weak direct field evidence	Not consistently observed; may cause permeability damage; often considered supplementary rather than primary mechanism.
Alkalinity effect	Sandstone and carbonates	Cation exchange ($Ca^{2+} \leftrightarrow H^{+}$); reactive crude oil acids; low-salinity injection	IFT reduction via in-situ soap formation; wettability alteration	Moderate laboratory support	Limited direct field-scale quantification	pH rise is often modest, not generally sufficient to significantly reduce IFT
Multicomponent ion exchange (MIE)	Sandstone (clay minerals); carbonates (with SO_4^{2-}, Ca^{2+}, Mg^{2+} as PDIs)	Presence of polar crude oil components; divalent cations; clay minerals (sandstone); potential determining ions (carbonate)	Wettability alteration via desorption of adsorbed oil	Strong laboratory and coreflood support	Supported by several pilot observations (indirect validation)	Requires specific ion composition; overlapping with EDL and wettability alteration mechanisms
Mineral dissolution	Carbonates > sandstones (anhydrite/calcite/ dolomite presence)	Reactive minerals; diluted brine; ionic strength variation; temperature sensitivity	Wettability alteration; porosity/permeability modification; pH increase	Moderate experimental evidence	Limited and case-dependent	Dissolution effects are not always observed; micro-CT studies show minimal structural change in some cases
Electrical double layer (EDL) expansion	Sandstone and carbonates	Reduction in ionic strength; presence of charged mineral surfaces; low multivalent ion concentration	Increased electrostatic repulsion; wettability alteration	Strong theoretical and lab support; mixed carbonate results	Limited direct field quantification	Difficult to isolate experimentally; contradictory findings in carbonate systems
Salt-in effect	Primarily sandstone (clay interaction)	Significant salinity reduction below the critical ionic strength	Increased solubility of polar organics; desorption from clay surfaces	Limited experimental support	No substantial field evidence	Poorly explored; lacks systematic correlation with mineralogy, pH, and ion type
Chemical osmosis	Oil-wet systems with high-salinity connate water	Semi-permeable membrane behaviour (oil film or clay layers); high-salinity gradient	Pressure-driven expansion of connate water; oil expulsion from dead-end pores	Conceptual and some lab-scale indications	Very limited field evidence	Requires an oil-wet system; does not explain ion-specific sensitivity; contribution likely minor
IFT reduction	Sandstone and carbonates	Presence of polar crude oil components; optimal ionic strength	Reduction in oil–water interfacial tension; capillary number increase	Mixed results; some lab support but many contradictory findings	Weak field-scale validation	IFT reduction is generally insufficient to explain observed recovery increments alone
Microdispersion formation	Sandstone (clay-rich) and carbonates	Presence of surface-active materials (asphaltenes, naphthenic acids); low-salinity dilution	Redistribution of polar components; wettability alteration; oil swelling	Strong micromodel and coreflood evidence	Emerging field relevance	Mechanistic overlap with EDL and osmosis; dependent on crude oil chemistry
Viscoelasticity	General (fluid–fluid interaction dominant)	Reduced ionic strength; presence of interfacial active components	Reduced snap-off; altered droplet trapping; improved displacement efficiency	Limited but growing experimental evidence	Not clearly validated at the field scale	Still emerging; contribution not fully quantified

algorithms has gained significant consideration for modelling and optimizing LSWF processes, particularly where conventional models are constrained by complex fluid–rock interactions and inherent multiscale reservoir heterogeneity. These algorithms enable the extraction of nonlinear relationships between reservoir properties, brine chemistry, and oil recovery. A wide range of ML techniques including artificial neural networks (ANN) (Hassoun, 1995), support vector machine (SVM) (Cortes and Vapnik, 1995), decision tree (DT), random forests (RF) (Breiman, 2001b), naive Bayes, k-nearest neighbours, gradient boosting (GB) (Chen and Guestrin, 2016), and linear regression, logistic regression (LR) have been employed to predict recovery factors and identify dominant controlling variables. In addition to predictive modelling, ML algorithms are increasingly used for sensitivity analysis and optimisation of injection strategies, including the identification of optimal salinity ranges and ion tuning (Jing et al., 2026). A wide range of supervised learning techniques, including ANNs, SVM, RFs, and GB methods, have been employed to predict recovery factors and identify dominant controlling variables. The strengths and weaknesses of each ML algorithm are presented in Table 3.

4.1.1. Artificial neural networks

Artificial neural networks (ANNs) have demonstrated significant proficiency in addressing complex engineering problems beyond the computational capacity of classical mathematics and conventional methodologies (Agwu et al., 2018). ANNs are sophisticated computational models designed to replicate the intricate structure and functional characteristics of biological neural networks in the human brain (Liang et al., 2023; Mansi et al., 2023). ANNs are widely used in LSWF research to model nonlinear relationships among injection water properties, reservoir lithology, and oil recovery. ANNs in LSWF applications require careful feature engineering to incorporate relevant variables, such as salinity, cation composition, clay fraction, porosity, and permeability. Data scaling and normalisation are critical due to the wide ranges of geological parameters. Additionally, ANNs are sensitive to hyperparameter selection, including the number of

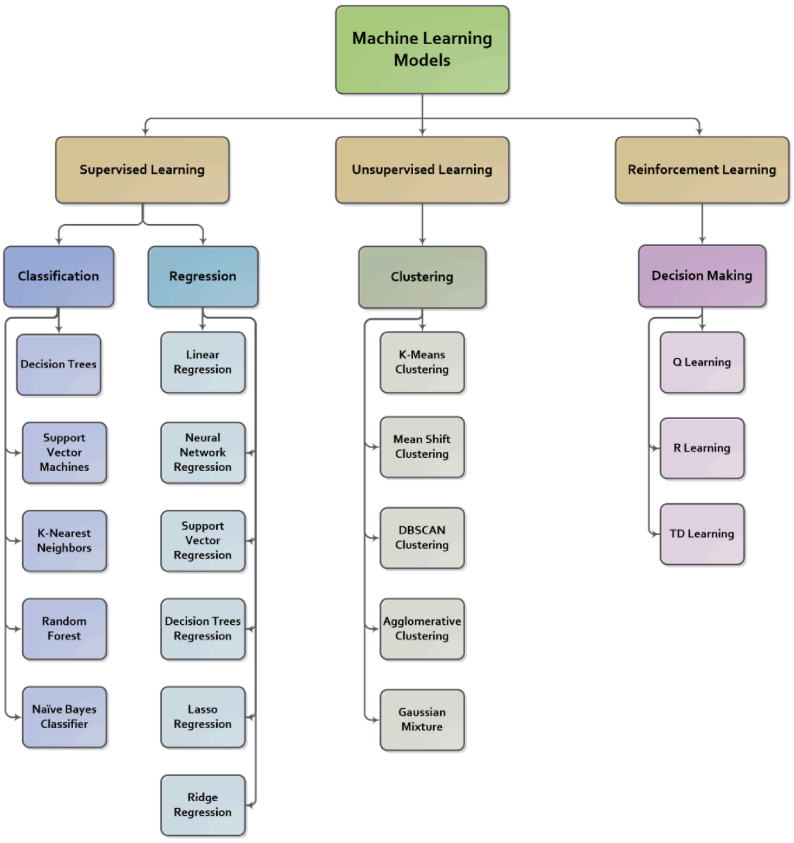


Fig. 10. Classification of machine learning models.

Table 3
Strengths and weaknesses of machine learning algorithms (Zhong et al., 2020, 2022).

Algorithm	Strengths	Weaknesses
ANN	• Robust to noise and missing values • Good performance on some tasks, such as image and text recognition	• Lack of interpretability • Requires data preparation • Computationally expensive
SVM	• Works well in complicated domains • Works well with outliers	• Requires data preparation • Kernel selection can be hard • Poor performance and long computation time if the dataset is large and noisy
RF	• No effort for data preparation • Able to rank feature importance • Works well in high-dimensional spaces	• Lack of interpretability if the tree quantity is large • May overfit if the data is noisy
XGBoost	• No effort for data preparation • Able to rank feature importance • State-of-the-art accuracy in many regression and classification problems	• Lack of interpretability • Poor performance on some tasks, such as image and text recognition

hidden layers, the number of neurons per layer, and the learning rate. However, despite their strong predictive performance, limited interpretability remains a key challenge. Therefore, techniques such as SHAP or sensitivity analysis are often employed to identify the dominant features driving recovery predictions.

4.1.2. Support vector machines

The support vector machine (SVMs) are a prevalent supervised learning algorithm introduced by Cortes and Vapnik (1995). SVM are a well-recognised technique widely used for classification and regression analysis. SVM have superior generalisation performance compared to other classification models in data mining (Zhang et al., 2017). SVM identifies a hyperplane that optimally separates a dataset into distinct classes. SVM uses a kernel technique to effectively map data into higher-dimensional spaces, facilitating improved separation of data points that are not linearly separable (Solano Meza et al., 2019). This mapping occurs within the current input domain in linear problems, whereas in nonlinear regression and classification problems, it occurs in multi-dimensional feature domains. SVMs have been applied to classify and regress LSWF outcomes, such as predicting wettability alteration or incremental oil recovery. The kernel choice (linear, RBF, polynomial) significantly impacts the model's ability to capture nonlinear interactions between brine chemistry and reservoir mineralogy. SVMs are particularly useful in small, high-dimensional datasets common in laboratory coreflood experiments. Class imbalance must be managed via oversampling or weighted penalties. Furthermore, hyperparameter tuning is critical to avoid overfitting, and SVM's interpretability is limited to the support vectors and sensitivity of the hyperplane.

4.1.3. Decision tree

The Decision Tree (DT) is a supervised learning technique commonly employed for classification tasks. In this method, ML models are constructed using graphical representations in the form of flow charts. This algorithm acquires the ability to make predictions by inferring rules from data characteristics (Shafiei et al., 2022). DT provides a transparent, white-box approach to identify key factors influencing LSWF performance. Their hierarchical structure allows mapping of complex rules, such as the combined influence of clay content, ionic composition, and rock type on oil recovery. DTs handle heterogeneous datasets well and are robust to missing data, but they are sensitive to small changes in the dataset, which can compromise generalization. Ensemble methods or pruning strategies are often applied in LSWF studies to mitigate overfitting.

4.1.4. Random forest

The random forest (RF) is a tree-based ensemble classifier (Breiman, 2001b). The RF algorithm operates on the principle of using multiple DT classifiers to both train the dataset and make predictions (Breiman, 2001a). The RF algorithm utilises an ensemble of trees that produce results autonomously. These results are then combined and analysed to determine the class that received the most votes, enabling the RF to make accurate predictions (Vasić et al., 2023). RF has been widely adopted in LSWF applications due to its robustness to noise, multicollinearity, and missing data. RFs are particularly suitable for high-dimensional reservoir datasets that contain mineralogical, petrophysical, and brine-chemistry features. Feature importance scores derived from RF models provide valuable insight into dominant parameters, e.g., the impact of divalent cations on incremental recovery. Additionally, proper tuning of the number of trees and depth is essential to balance accuracy and computational efficiency.

4.1.5. Gradient boosting

The Gradient Boosting (GB) regressor is an ensemble model consisting of a series of trees trained sequentially. Each subsequent model in the sequence learns from the mistakes made by the previous model. This AI regressor uses the boosting approach to augment the precision of its predictions. GB methods have been employed to capture subtle, nonlinear interactions in LSWF datasets. GB is effective in predicting incremental oil recovery under varying salinity conditions. However, careful attention to the learning rate and tree depth is required to prevent overfitting, especially when datasets are small, as is often the case in coreflood experiments.

4.1.6. Naive Bayes

The Naïve Bayes (NB) classifier is a supervised machine learning algorithm that uses statistical methods to estimate the likelihood that an instance belongs to a given class. The algorithm is a simple but very effective method for solving both binary and multiple-class classification problems. The NB demonstrates exceptional accuracy and speed when applied to large datasets, while also performing very well on smaller datasets (Rajkumar and Ganapathy, 2020). Although less common in LSWF research, Naïve Bayes classifiers can efficiently handle probabilistic predictions for categorical outcomes, such as wettability type or recovery classification. Their assumption of feature independence is often violated in geochemical and petrophysical datasets; however, it provides a fast, interpretable baseline for comparison with more complex models.

4.1.7. Bagging classifier

The Bagging classifier (BC), also known as bootstrap aggregation, is an ML algorithm that trains each base classifier on a randomly redistributed version of the original training dataset. Subsequently, the classifiers' forecasts are combined using averaging or voting techniques to produce a final prediction (Breiman, 1996). In LSWF applications, BC reduces variance and improves prediction stability in heterogeneous datasets. This helps integrate multiple experimental cores or simulation outputs, enhancing robustness against noisy or imbalanced datasets and helping prevent overfitting.

4.1.8. Logistic regression

Logistic regression (LR) is a statistical technique used to estimate the probability of a binary result. LR analyses data and elucidates the association between a single binary dependent variable and one or more independent variables, which may be nominal, ordinal, interval, or ratio-level (Patel and Prajapati, 2018). LR is renowned for its parallelizability, simplicity, and interpretability (Kardani et al., 2021). In LSWF, LR is commonly used for binary classification tasks, such as predicting the effect of low-salinity water on oil recovery in a given core sample. LR is valued for its interpretability, enabling the identification of key predictors such as cation exchange capacity or clay fraction, though it may struggle with complex nonlinear interactions without preprocessing or interaction terms.

4.1.9. Extremely randomised tree

The Extremely Randomised Tree (ET) technique is a recently proposed novel tree-based approach to supervised classification. This technique was designed to reduce the probability of overfitting a dataset. ET employs a methodology similar to that of an RF, where each base estimator is trained using a random subset of features (John et al., 2016). However, it partitions the node by randomly selecting the optimal features and corresponding values. ET uses the entire training dataset to train each regression tree. Conversely, RF trains the model using a bootstrap replica (Geurts et al., 2006). In LSWF modelling, ETs are useful for ranking feature importance and capturing subtle trends in small or noisy laboratory datasets, making them suitable for high-throughput screening of salinity effects across multiple rock types.

4.2. Application of machine learning in low-salinity waterflooding

Optimizing LSWF techniques can pose significant computational challenges, particularly when the reservoir models being studied are complex. The proliferation of ML techniques has significantly addressed the computational challenge, as evidenced by numerous scholarly works. In the petroleum sector, ML methods are widely used to analyse the enormous quantity of experimental and field data collected to predict and optimize field performance. However, a limited number of ML models have been employed to evaluate the effectiveness of LSWF in improving oil recovery using intelligent prediction models. Table 4 provides a concise summary of ML applications in LSWF for improved oil recovery.

Recently, Kandala and Govindarajan (2025) developed a comprehensive ML-based surrogate modelling framework for predicting oil recovery during tertiary LSWF in Berea sandstone. They used 10,240 simulation-generated samples based on a physics-informed MIE mechanism. Four key ionic inputs (Na^+, Ca^{2+}, Mg^{2+}, Cl^-) were systematically varied, and eight ML algorithms were benchmarked with hyperparameter tuning via GridSearchCV and five-fold cross-validation. Models were trained separately for each flooding time step (200–400 days) rather than

Table 4
Summary of recent studies on the application of ML in LSWF.

Ref.	Reservoir type	ML models	Key input features	Dataset size	Train/Test split	Validation metrics	Performance (best Model)
Kandala and Govindarajan (2025)	Sandstone	RF, ANN, DT, GB, LR, RR	Injection water ion concentrations (Na^+, Ca^{2+}, Mg^{2+}, Cl^-); flooding time	10,240 simulated data points	75% train/25% test	R^2, MAE, RMSE, MSE	Best: RF $R^2 \approx 0.995$, MAE ≈ 0.001 Consistent across 200–400 days
(Shafiei et al., 2022)	Carbonate	ANN, SVM, DT, RF, CMIS	Oil properties; rock properties; connate & injected water composition; operational parameters	582 data points (from 47 studies)	80% train/20% test	R^2, RMSE, SD, MRD, MARD	Best: RF RMSE (train) 2.497 RMSE (test) 5.757 R^2 (test) 0.840 MARD (test) 3.302%
Hidayat and Astsauri (2022)	Carbonate	RF regression	Injection water composition; formation water composition; injected pore volume; reservoir temperature; injected brine temperature	1000 synthetic simulation runs	Multiple random states	R^2, MSE, MAE	Best: RF: $R^2 = 0.935$ (inflated)
Kondori et al. (2021)	Sandstone & carbonate	LSSVM-CSA, ANFIS, ET	Brine salinity; injection rate; oil viscosity; temperature; porosity; permeability	257 data points	90% train/10% test	R^2, RE, MAPE, MSE	Best: LSSVM-CSA $R^2 = 0.988$ (total) MAPE = 1.29% (train) MSE = 1.43 (total)
Kalam et al. (2021)	Heterogeneous carbonate	ANN	Water cut; permeability variation; mobility ratio; dilution coefficient	20,000 simulated data points	70% train/30% test	R^2, RMSE	Best: ANN with LM algorithm (25 neurons) $R^2 = 0.9989$ (train)/0.998 (test) AAPE = 1.70% (train)/1.84% (test) RMSE = 0.0061 (train)/ 0.0086 (test)
Tatar et al. (2021)	Sandstone	ANN, SVM, LR, CMIS	Na^+ (injected brine); Ca^{2+} (connate water); TAN; porosity; operational factors	1316 data points (99 studies)	80% train/20% test;	RMSE, MARD	Best: CMIS $R^2 = 0.901$ (train)/0.731 (test) RMSE = 4.62 (train)/7.76 (test) MARD = 2.81% (train)/ 5.27% (test) MRD = 0.072% (test)
Salimova et al. (2021)	Carbonate/ Limestone	DT, ANN, SVM	Oil/brine parameters (nonlinear relationships emphasized)	5000 simulated data points	70% train/30% test;	Correlation analysis	Best: DT $R = 0.68$ (train)/0.63 (test)
Wang and Fu (2018)	Sandstone	MLR, ANN, SVM, RF	Salinity; rock properties; brine characteristics	202 tertiary RF groups; 157 secondary RF groups (from 200+ publications)	80% train/20% test	R^2	Best: RF $R^2 = 0.728$ (average) SVM: Best for brine–rock correlation MLR adj. $R^2 \leq 0.139$

treating time as a continuous feature, to account for the evolving nonlinearity of the recovery process. The RT model achieved outstanding performance across all flooding periods, with $R^2 \approx 0.995$ and MAE ≈0.001, as shown in Fig. 11, demonstrating high accuracy, stability, and unbiased residual behaviour. The fifth-order Polynomial Regression and Gradient Boosting models also performed well, with $R^2 > 0.97$, while linear models were inadequate because they could not capture the nonlinear ion–recovery relationships. The study relies on synthetic data generated from a single, homogeneous Berea sandstone model, which limits generalizability to real, heterogeneous reservoirs. Additionally, the exclusion of key reservoir and fluid properties (e.g., wettability, mineralogy, oil composition) limits physical completeness.

Shafiei et al. (2022) developed ML predictive models, including ANN, SVM, DT, RT, and CMIS, to forecast the recovery factor of LSWF in carbonate reservoirs. The authors used a database of 582 data points assembled from 47 experimental coreflooding studies. A comprehensive preprocessing workflow was implemented, including duplicate removal, mean imputation, VIF-based collinearity control, outlier removal, and feature scaling. The RF achieved the best predictive performance, with RMSEs of 2.497 and 5.757 for the training and test datasets, respectively, and R^2 values of 0.9722 and 0.8399, as illustrated in Fig. 12. Williams plot applicability domain analysis confirmed database reliability, with zero training outliers and only nine testing outliers. However, the dataset used in the study is exclusively carbonate-focused, limiting generalizability to sandstone or mixed-lithology systems. The key oil property features, including TAN, TBN, saturates, aromatics, resins, and asphaltenes, were largely excluded, potentially underrepresenting crude oil compositional effects on LSWF mechanisms. The DT model exhibited clear overfitting with RMSE values of 0.293 and 7.117 for training and testing, respectively. While the CMIS ensemble integrated only ANN and SVM outputs, excluding RF and DT, which may have constrained its predictive ceiling.

In another study, Hidayat and Astsauri (2022) applied an RF regression algorithm for parameter sensitivity analysis of LSWF in carbonate reservoirs. The primary objective of the study was to identify the most influential parameters governing oil recovery during the LSWF process. The dataset was generated synthetically using 1000 Design of Experiments (DoE) simulations based on a CMOST reservoir model. The RF model was benchmarked against Multivariate Linear Regression (MLR), ANN, SVM, K-Nearest Neighbours, and DT, demonstrating superior and consistent

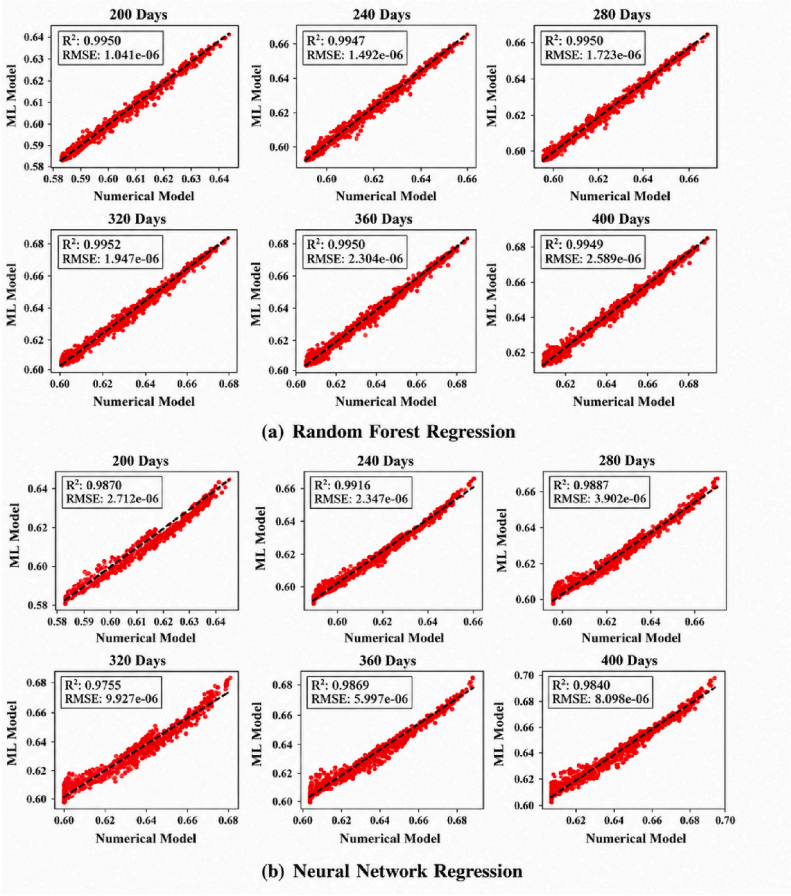


Fig. 11. Comparison of surrogate and numerical model oil recovery factor predictions: (a) Random Forest and (b) Neural Network (Kandala and Govindarajan, 2025).

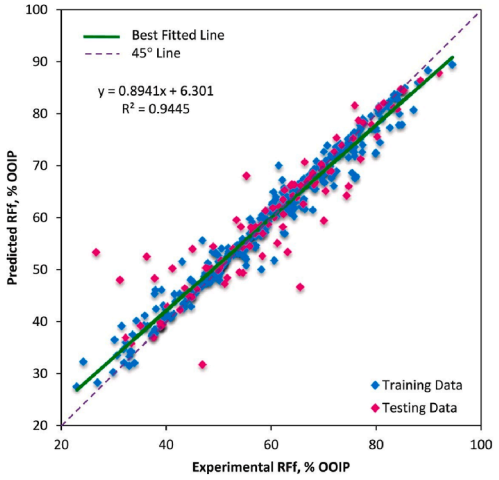


Fig. 12. Cross-plot of predicted versus experimental final recovery factor values for the RF model across training and testing datasets (Shafiei et al., 2022).

performance, with R^2 exceeding 0.9 across all five random-state variations for both the training and test datasets, as shown in Fig. 13. However, the study uses dataset generated entirely from simulation, raising concerns about its ability to represent the complexity of real carbonate reservoirs. While the reliance on a single conceptual model limits generalizability across diverse lithologies and fluid systems. Additionally, the negligible role of temperature contradicts experimental evidence, suggesting inadequate representation of temperature-dependent geochemical processes.

Kondori et al. (2021) assessed the efficacy of LSWF using an adaptive network-based fuzzy inference system (ANFIS) coupled with simulated annealing (CSA), the hybridized least squares support vector machine (LSSVM), and other algorithms. They compiled a dataset of 257 LSWF data points from 27 experimental and simulation studies covering both sandstone and carbonate reservoirs. Among the evaluated models, the LSSVM-CSA model achieved the best performance (test $R^2 = 0.976$, MAPE = 1.83%), demonstrating strong predictive capability and stable convergence. Whereas, feature importance analysis identified TDS and oil viscosity as dominant variables, aligning well with established LSWF mechanisms such as wettability alteration and fluid mobility control, while injection rate showed minimal influence, offering useful operational insights. However, using a single 90/10 train–test split yields a very small test set (approximately 26 samples), making performance metrics statistically unstable and sensitive to data partitioning. Overfitting is observed in the ET model, with an extremely high training R^2 of 0.995 (Fig. 14), due to the model's high complexity relative to the dataset size. Additionally, the exclusion of key physicochemical parameters (e.g., pH, clay content, and fines migration) due to data scarcity restricts mechanistic completeness.

Furthermore, Kalam et al. (2021) developed a large-scale ML framework for LSWF using 20,000 synthetically generated data points derived from a calibrated black-oil reservoir simulator based on the Madison Formation carbonate reservoir in Wyoming. A comprehensive ANN optimisation study identified the

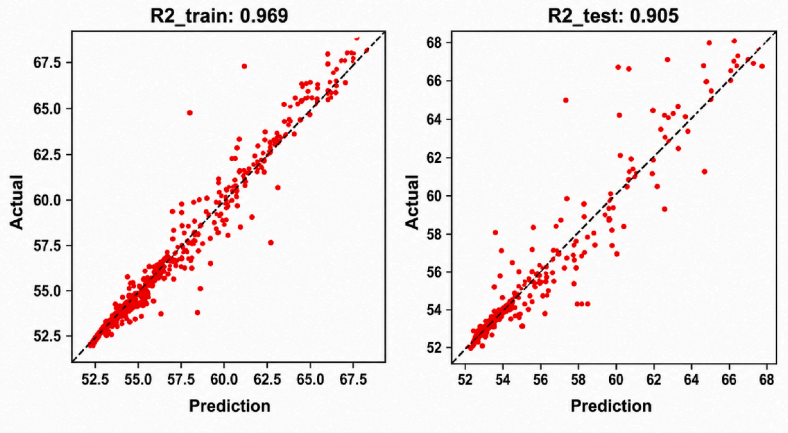


Fig. 13. The actual vs. predicted plot for evaluating model performance (Hidayat and Astsauri, 2022).

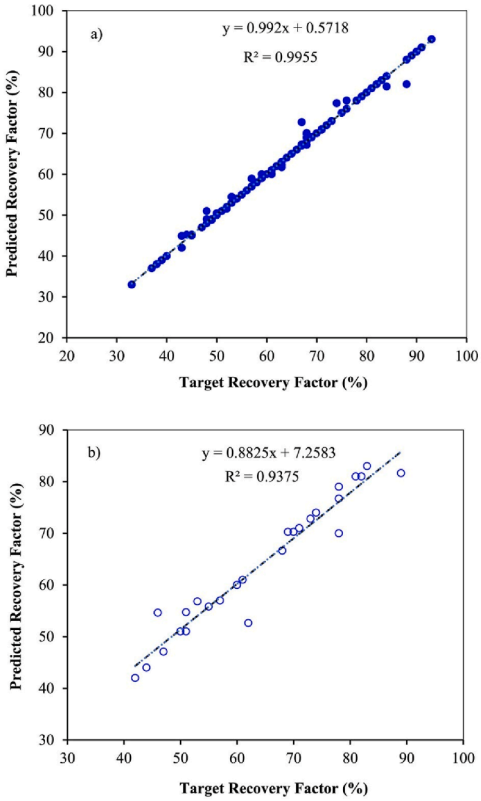


Fig. 14. ET model performance to estimate recovery factor of LSWF process: a) training and b) testing phases (Kondori et al., 2021).

Levenberg–Marquardt algorithm with 25 hidden neurons and tan-sigmoid activation as optimal, achieving excellent predictive performance (test $R^2 = 0.998$, AAPE $\approx$ 1.84%) with negligible overfitting due to the large, noise-free dataset, as illustrated in Fig. 15. The model is fully reproducible, presented as an explicit mathematical correlation with published weights and biases, and is particularly valuable for rapid screening and sensitivity analysis of operational parameters such as dilution ratio, mobility ratio, and heterogeneity. However, the reported performance reflects surrogate accuracy against simulator outputs rather than true physical prediction capability. The dataset is entirely synthetic and derived from a single reservoir model, limiting generalizability to other formations. Furthermore, the critical physicochemical factors governing LSWF mechanisms, such as brine ionic composition, crude oil chemistry, and rock mineralogy, are not included, restricting mechanistic interpretability.

Additionally, Tatar et al. (2021) developed the most comprehensive and rigorously preprocessed LSWF ML dataset, comprising 1316 data points from 99 sandstone coreflood studies. A structured six-stage preprocessing workflow, encompassing duplicate removal, variance filtering, missing data treatment, collinearity reduction, outlier detection, and recursive feature elimination, reduced the dataset to 11 key features representing rock, fluid, and operational parameters. Four models, including LR, MLP, SVM, and the CMIS ensemble, were trained; the CMIS ensemble achieved the best performance, with $R^2 = 0.731$ and RMSE = 7.76% OOIP, as shown in Fig. 16. The near-zero MRD (0.072%) confirmed unbiased predictions, while cross-validation and applicability domain analysis demonstrated strong generalisation with minimal overfitting. Despite these strengths, limitations remain, particularly related to dataset heterogeneity and feature dominance (e.g., initial recovery factor), which may bias model learning toward secondary recovery behaviour rather than isolating LSWF effects. Furthermore, during feature analysis, the key elements influencing LSWF in sandstone rocks were found to be the Na + concentration in the injected brine, the Ca^{2+} concentration in the connate water, TAN, and porosity.

Finally, Wang and Fu (2018) conducted the first data-driven analyses of LSWF by compiling experimental results from over 200 publications, resulting in approximately 200 tertiary recovery data points. They developed multiple predictive models, including ANN, SVM, RF, and MLR, to systematically evaluate their performance across varying input parameters and oil recovery conditions, to identify the most effective modelling approach. Their analysis demonstrated that single-parameter correlations are weak (maximum R ~ 0.26), while multivariable linear models explain only a small fraction of the variance (adjusted $R^2 \leq 0.13$), confirming the strongly nonlinear, multi-parameter nature of LSWF. Among ML models, RF achieved the best performance (R^2 ~ 0.73), outperforming ANN and SVM. Most importantly, the study established that no single parameter governs LSWF effectiveness and identified the crude oil base number as the most influential variable, supporting mechanistic interpretations of wettability alteration. However, the compiled dataset is highly heterogeneous due to multiple laboratories, leading to inconsistencies in experimental conditions and measurements. The target variable is strongly imbalanced and right-skewed, biasing models toward low recovery predictions, yet no imbalance

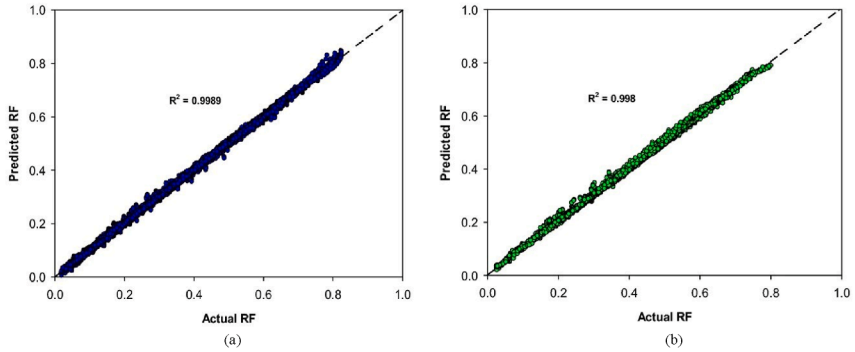


Fig. 15. Predicted vs actual RF scatter plots for (a) training and (b) testing datasets (Kalam et al., 2021).

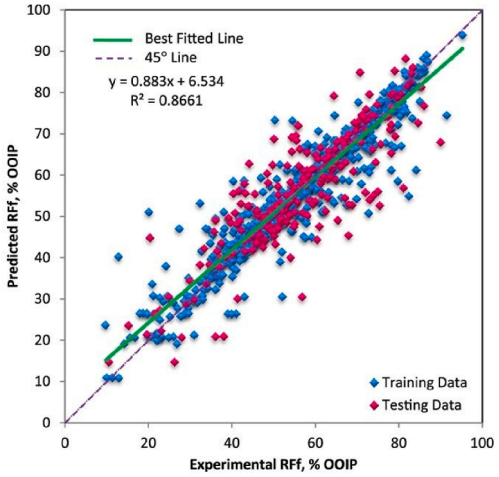


Fig. 16. Cross-plot of predicted and experimental final recovery factor values for the CMIS model across training and testing datasets (Tatar et al., 2021).

mitigation strategies were applied. The small test set, with approximately 36 samples from a single train–test split, yields statistically unstable performance estimates, with no cross-validation or uncertainty quantification. Additionally, ML models were trained on different data subsets due to missing values, limiting fair comparison, and no trained model parameters were reported, restricting reproducibility.

5. Challenges and limitations

The integration of ML has significantly improved the efficiency and predictability of enhanced oil recovery (EOR) processes. However, its implementation in LSWF remains subject to several challenges. While ML techniques offer powerful tools for pattern recognition and nonlinear prediction, various limitations constrain their effectiveness in LSWF.

- A primary limitation in LSWF is the scarcity of large, high-quality field-scale datasets (Sircar et al., 2021). Most ML models developed for LSWF are trained using laboratory core-flood experiments or simulation data rather than full-field operational data. The core-scale experiments provide controlled environments for studying LSWF mechanisms, and do not fully capture reservoir-scale heterogeneity, geochemical evolution, and dynamic flow interactions. Consequently, models trained on laboratory data may suffer from limited generalizability when applied to field conditions.
- Another significant issue relates to feature variability and data inconsistency, which can compromise the reliability of predictive models (Xu et al., 2022). LSWF performance depends on a complex combination of various parameters, including brine composition (e.g., salinity, divalent ion concentration), crude oil chemistry (e.g., TAN, polarity), rock mineralogy, porosity, permeability, temperature, and operational parameters. However, these variables are not consistently measured or reported across studies. Differences in experimental protocols, aging procedures, brine preparation methods, and measurement techniques introduce variability that can degrade model reliability.
- Overfitting is particularly problematic in LSWF applications due to limited datasets (Tariq et al., 2021). Many published models report high predictive accuracy (e.g., $R^2 > 0.99$), yet these results are often derived from relatively small datasets or narrow operational ranges. Without rigorous cross-validation or external validation on independent datasets, model performance may be overly optimistic. The lack of standardised benchmarking datasets for LSWF further limits objective comparison among ML approaches.
- Uncertainty quantification represents another critical limitation in LSWF modelling, affecting the reliability of predictive outcomes (Almeida and Rochinha, 2023). Most ML studies in LSWF focus primarily on deterministic recovery-factor prediction, without explicitly accounting for prediction uncertainty. Given the economic and operational risks associated with EOR implementation, uncertainty-aware models such as probabilistic ML frameworks or Bayesian approaches are essential for risk-informed decision-making. Integrating uncertainty quantification into ML-based LSWF screening and optimisation remains an underexplored research direction.
- Transfer learning is another promising and emerging data-driven approach (Jiang et al., 2023), but is largely untapped in LSWF modelling, where limited field-scale data can be complemented by pre-trained models derived from analogous reservoirs or synthetic datasets to enhance predictive robustness. However, systematic frameworks for transferring knowledge across lithologies (e.g., sandstone to carbonate) or across salinity regimes have not yet been fully established.
- The computational cost and model interpretability remain practical concerns in LSWF applications. Advanced deep learning architectures may improve predictive capability but often operate as “black-box” systems, limiting mechanistic insight. In LSWF, where understanding the interplay between geochemical reactions and flow behaviour is essential, model transparency is particularly important. Therefore, hybrid approaches that integrate physics-based understanding with data-driven modelling may provide a more reliable pathway forward.

6. Recommendation for future work

To improve the robustness and applicability of ML in LSWF, future studies should focus on addressing the following:

- Developing a deeper understanding of the coupled physico-chemical mechanisms governing rock–fluid interactions, particularly under varying reservoir conditions such as temperature, pressure, and mineralogical heterogeneity. Advanced experimental techniques, combined with pore-scale imaging and geochemical modelling, could provide valuable insights into the pore-scale processes responsible for wettability alteration and fines migration.
- The integration of physics-based reservoir simulation with ML offers a promising approach for predicting LSWF performance. The hybrid models can enhance predictive reliability in LSWF by combining a data-driven framework with fundamental physical principles, enabling accurate forecasts while preserving mechanistic interpretability and reducing reliance on large empirical datasets.
- The development of digital rock-physics workflows, automated salinity-optimisation algorithms, and real-time adaptive injection strategies represents a transformative direction for LSWF deployment. The integration of multi-scale data from laboratory experiments, core flooding tests, and field pilots into unified ML platforms could facilitate the development of robust, reservoir-specific screening and design tools for smart water injection.
- Advancing LSWF from a largely empirical technique toward an intelligent, autonomous IOR/EOR system requires interdisciplinary collaboration between reservoir engineers, geochemists, and data scientists. The convergence of mechanistic insight and ML-driven optimisation will define the next generation of IOR technologies.
- Finally, more field-scale pilot studies and long-term monitoring programs are required to validate laboratory findings and improve the reliability of predictive models. Moreover, strengthened collaboration between academia and industry will be essential to establish comprehensive datasets and accelerate the practical deployment of LSWF technologies across diverse reservoir environments.

7. Conclusion

- This work provides a comprehensive review of low-salinity waterflooding (LSWF), highlighting the fundamental physico-chemical mechanisms responsible for improved oil recovery, including electric double-layer expansion (ELD), multicomponent ion exchange (MIE), mineral dissolution, and microdispersion.
- The review systematically examines the role of machine learning (ML) techniques in LSWF applications, summarising recent developments in predictive modelling for recovery factor estimation, reservoir screening, and injection parameter optimisation.
- A comparative assessment of commonly used ML algorithms is presented, highlighting their predictive capabilities and advantages over conventional modelling techniques.
- The study identifies key challenges associated with ML-driven LSWF applications, including data availability, model interpretability, uncertainty quantification, and integration with real-time reservoir data.
- Finally, the review outlines future research directions by emphasising the potential of hybrid physics-informed ML models, digital rock physics workflows, and data-driven optimisation frameworks to enable intelligent, adaptive LSWF strategies for improved oil recovery.

CRediT authorship contribution statement

Aftab Hussain Arain: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Ahmad Salam Farooqi:** Writing – review & editing. **Rayed S. Alshareef:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Acronyms

ANN	Artificial Neural Network
CEC	Cation Exchange Capacity
COBR	Crude oil/Brine/Rock
CMIS	Clustering for Multiple Instance Selection
CSW	Controlled Salinity Water Flooding
DT	Decision Tree
EOR	Enhanced Oil Recovery
FTIR	Fourier Transform Infrared
GB	Gradient Boosting
IFT	Interfacial Tension
IOR	Improved Oil Recovery
LR	Logistic Regression
LSWF	Low-Salinity Waterflooding
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MLR	Multi Linear Regression
MSE	Mean Square Error
HSWF	High-Salinity Waterflooding
OOIP	Original Oil in Place
RF	Random Forest
RMSE	Root Mean Square Error
SVM	Support Vector Machine
SAM	Surface Active Materials
TAN	Total acid Number
TDS	Total Dissolved Solids
XGB	Extreme Gradient Boosting

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